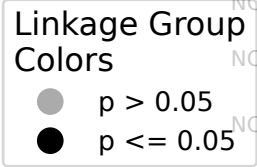


diamond: 10279 orthologs on 17 Mare scaffolds and 12 Cgig scaffolds.



	Mare scaf	Cgig scaf	p (FET)	Count
0	NC_069132.1	NC_088858.1	0.00e+00	781
1	NC_069123.1	NC_088859.1	0.00e+00	776
2	NC_069134.1	NC_088859.1	0.00e+00	685
3	NC_069129.1	NC_088855.1	0.00e+00	604
4	NC_069135.1	NC_088856.1	0.00e+00	541
5	NC_069125.1	NC_088855.1	9.43e-142	534
6	NC_069124.1	NC_088854.1	1.02e-159	516
7	NC_069125.1	NC_088857.1	4.11e-176	487
8	NC_069124.1	NC_088858.1	5.91e-140	487
9	NC_069133.1	NC_088860.1	0.00e+00	462
10	NC_069138.1	NC_088854.1	1.63e-227	359
11	NC_069127.1	NC_088861.1	0.00e+00	358
12	NC_069130.1	NC_088855.1	2.74e-187	336
13	NC_069136.1	NC_088860.1	4.63e-287	330
14	NC_069122.1	NC_088862.1	0.00e+00	296
15	NC_069131.1	NC_088857.1	9.20e-196	285
16	NC_069126.1	NC_088857.1	9.16e-91	248
17	NC_069128.1	NC_088854.1	3.83e-90	201
18	NC_069126.1	NC_088854.1	1.28e-35	196
19	NC_069137.1	NC_088853.1	2.05e-110	93
20	NC_069128.1	NC_088853.1	2.30e-05	28

	Mare scaf	Cgig scaf	p (FET)	Count
0	NC_069132.1: 0-end	NC_08858.1: 0-end	0.00e+00	781
1	NC_069123.1: 0-end	NC_08859.1: 0-end	0.00e+00	776
2	NC_069134.1: 0-end	NC_08859.1: 0-end	0.00e+00	685
3	NC_069129.1: 0-end	NC_08855.1: 0-end	0.00e+00	604
4	NC_069135.1: 0-end	NC_08856.1: 0-end	0.00e+00	541
5	NC_069125.1: 0-end	NC_08855.1: 0-end	9.43e-142	534
6	NC_069124.1: 0-end	NC_08854.1: 0-end	1.02e-159	516
7	NC_069125.1: 0-end	NC_08857.1: 0-end	4.11e-176	487
8	NC_069124.1: 0-end	NC_08858.1: 0-end	5.91e-140	487
9	NC_069133.1: 0-end	NC_08860.1: 0-end	0.00e+00	462
10	NC_069138.1: 0-end	NC_08854.1: 0-end	1.63e-227	359
11	NC_069127.1: 0-end	NC_08861.1: 0-end	0.00e+00	358
12	NC_069130.1: 0-end	NC_08855.1: 0-end	2.74e-187	336
13	NC_069136.1: 0-end	NC_08860.1: 0-end	4.63e-287	330
14	NC_069122.1: 0-end	NC_08862.1: 0-end	0.00e+00	296
15	NC_069131.1: 0-end	NC_08857.1: 0-end	9.20e-196	285
16	NC_069126.1: 0-end	NC_08857.1: 0-end	9.16e-91	248
17	NC_069128.1: 0-end	NC_08854.1: 0-end	3.83e-90	201
18	NC_069126.1: 0-end	NC_08854.1: 0-end	1.28e-35	196
19	NC_069137.1: 0-end	NC_08853.1: 0-end	2.05e-110	93
20	NC_069128.1: 0-end	NC_08853.1: 0-end	2.30e-05	28

(a) depicts the chromosomes/scaffolds of Mare (x-axis) plotted against the chromosomes/scaffolds of Cgig (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10279 orthologs found between 17 Mare scaffolds and 12 Cgig scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.